

1 June 2026

To the Editor-in-Chief

Experimental Mathematics

Dear Editor,

I am pleased to submit the manuscript “**Optimal strategies in the all-heads coin game**” for consideration as a research article in *Experimental Mathematics*.

Summary of the contribution. The paper analyses a sequential coin-flipping game with two parameters: p , the probability of showing heads, and n , the number of coins. We give the winning probability and the optimal strategy within that game for $p \geq 1/2$ or $p = \frac{1}{2} - \delta$ for small $\delta > 0$.

Fit with the journal. The work is experimental mathematics in a concrete sense. The perturbative results for $p < 1/2$ were found by arbitrary-precision computation of the Bellman recursion before they were proved. In addition, every numbered theorem, proposition, lemma, and corollary of the paper has been formally verified in Lean 4 / Mathlib.

On refereeing the formalization. Although EM has several experts in Lean/Mathlib on its editorial board, the formalization is structured so that it can be assessed by a referee who is *not* a Lean expert (as we hope; see Appendix A).

Code and data availability. The Lean 4 / Mathlib formalization and the numerical code supporting the paper are openly available in the public repository *coins* (<https://github.com/pfaffelh/coins>) and are permanently archived on Zenodo with DOI 10.5281/zenodo.20489409, as stated in the manuscript’s data availability statement.

Disclosure on the use of AI tools. In the interest of full transparency, the manuscript states openly (in the note on authorship in Section 1 and the development history in Appendix A) that the drafting of prose, the Lean proof tactics, and the numerical scripts were produced with Anthropic’s Claude under my direction. Of course, I have reviewed every line and take full responsibility for the manuscript.

Thank you for considering this submission. I look forward to hearing from you.

Sincerely,

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